

Dominant vs. Dominated: Concept-Level Generative Collapse in Diffusion Models

Anonymous ECCV 2026 Submission

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Abstract. Text-to-image diffusion models have attracted significant attention for their ability to generate diverse, high-fidelity images. However, in multi-concept generation, one concept token often dominates the output while others are suppressed—a phenomenon we term the Dominant-vs-Dominated (DvD) imbalance. To systematically study this failure mode, we introduce DominanceBench and examine its underlying causes from both data and internal-mechanistic perspectives. Our controlled fine-tuning study, which mimics concept learning during diffusion-model training, shows that concepts learned from visually homogeneous (low-variation) concept-specific training images exhibit stronger dominance when composed with others. Cross-attention analysis indicates that dominant tokens concentrate attention in early denoising steps, followed by reduced representation of competing concepts. Head-ablation analysis further shows that this dominance is distributed across attention heads rather than localized. Overall, these findings characterize DvD as a systematic concept-level failure mode and provide a basis for more reliable and controllable multi-concept generation. DominanceBench will be released upon publication.

Keywords: Text-to-Image Generation · Multi-Concept Generation · Diffusion Models

1 Introduction

Text-to-image diffusion models [12, 18, 28, 32, 34, 37, 51] have achieved remarkable success in generating high-quality images from textual descriptions. However, ensuring the model’s representational fidelity to textual concepts [50] remains a fundamental challenge. Recent research has extensively explored this limitation from complementary perspectives. One line of work investigates *memorization* [5, 7, 14, 17, 21, 36, 38, 42, 43, 48], where models reproduce near-identical images across different random seeds, primarily due to excessive duplication of specific image-prompt pairs in training data. Another line studies compositional text-to-image generation, aiming to reduce concept interference and binding errors under multi-concept prompts [3, 10, 13, 30].

In this work, we examine a complementary aspect that arises from the *interplay* of these two dimensions—training data characteristics and multi-concept compositional capability. Specifically, we focus on visual dominance in multi-concept generation: one concept visually dominates the generation while the

“Neuschwanstein Castle *coaster*”

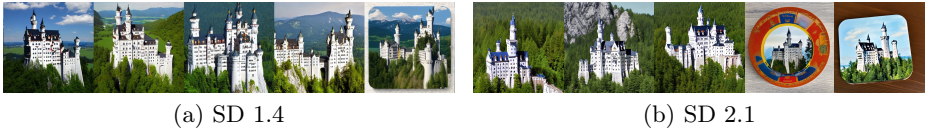


Fig. 1: Generation results across five random seeds. Only one (SD 1.4) and two (SD 2.1) out of five seeds successfully generate both concepts.

other is suppressed and rarely appears. For example, as shown in Fig. 1, we generate images with Stable Diffusion (SD) v1.4 and v2.1 using the prompt “Neuschwanstein Castle coaster” using different random seeds. The castle concept is consistently presented in the generated images, whereas the coaster concept rarely appears. We refer to this as the *Dominant-vs-Dominated (DvD) phenomenon*.

Motivated by these observations, we study DvD as a concept-level imbalance induced by visual dominance between concepts. Unlike memorization, which concerns prompt-specific reproduction, and prior work that aims to improve compositional text-to-image generation under multi-concept prompts, we hypothesize that limited variation in a concept’s training images (visual homogeneity) can suppress other concepts during multi-concept generation. We conjecture that it may do so by inducing relatively rigid visual priors that dominate compositions with more diverse concepts.

To causally test our hypothesis, we conduct controlled fine-tuning studies that simulate concept learning: we learn new concepts from curated training image sets while systematically varying within-concept visual diversity, and observe that dominance increases monotonically as diversity decreases. To reveal the internal mechanisms of DvD, we analyze cross-attention dynamics and head ablations, revealing that (1) DvD prompts exhibit significantly higher attention concentration on dominant tokens in lower-spatial-resolution layers in the U-Net (UNet) at early denoising steps, (2) dominated concepts experience rapid attention decay in the early phase of the denoising process, and (3) unlike memorization which is localized to specific heads, DvD emerges from distributed cooperation among multiple attention heads. To support this mechanistic analysis, we curate DominanceBench (Sec. 3.3), a benchmark of 300 two-concept prompts exhibiting strong DvD behavior.

Our main contributions are:

- We characterize the DvD phenomenon and curate DominanceBench for its systematic evaluation.
- We demonstrate through controlled fine-tuning studies on newly learned concepts that limited variation in concept-specific training images plays a causal role in DvD.
- We reveal the internal mechanisms of DvD through cross-attention analysis and head ablations, identifying where (lower-spatial-resolution layers), when (early timesteps), and how (distributed across heads) dominance manifests during generation.

2 Related Work

Text-to-image diffusion models have rapidly advanced in fidelity and controllability [12, 18, 28, 32, 34, 37, 51], with surveys summarizing this progress [4, 49, 50]. Two research directions are especially relevant: memorization in diffusion models, and compositional text-to-image generation under multi-concept prompts. Our work connects these directions by studying concept-level dominance in multi-concept generation and its relationship to training data properties.

2.1 Memorization in Diffusion Models

Memorization refers to the reproduction of near-identical training images. It has been studied from both empirical and mechanistic perspectives [5, 42, 43]. Prior work analyzes memorization signals in cross-attention patterns and token-level attention [7, 36], in prediction magnitudes and detection criteria [48], and in localized internal units such as neurons [14]. Complementary theoretical lenses connect memorization to geometry and sharpness of learned distributions [17, 21, 38]. We use memorization as a reference point when interpreting DvD mechanisms.

2.2 Multi-Concept Generation and Evaluation

Multi-concept prompts pose persistent challenges for text-to-image diffusion models, including concept interference, binding errors, and missing subjects. A broad line of work seeks to improve compositional generation and editing via attention control and guidance—for example, through attention-based semantic steering [6], cross-attention control for edits [3, 8, 13], attention-map alignment and control [35, 46], and test-time attention segregation [1]. Other approaches address compositionality through model- and objective-level designs, including composable diffusion and energy-based variants [11, 22, 26, 44, 53], as well as VLM-based feedback to better satisfy complex prompts [47].

Multi-concept customization further stresses compositional behavior when combining user-specific concepts [10, 19, 23, 24, 27, 30, 39]. In parallel, benchmarks and evaluator studies aim to measure compositional fidelity and diagnose failure modes at scale, often using pretrained VLMs as automated judges [9, 15, 16, 20, 40, 45]. Building on these advances in compositional generation and evaluation, we take a data-centric perspective and investigate an upstream driver of systematic dominance: concept-level visual homogeneity in training images.

3 The Dominant-vs-Dominated Phenomenon

3.1 Phenomenon Definition

Prior work studies memorization—reproducing near-identical training examples from duplicated image-text pairs [5, 43, 48]—and methods that improve multi-concept compositional generation [3, 10, 13, 30]. Here we connect these perspectives by examining an upstream data property—concept-level visual homogeneity in training images—that can induce systematic visual dominance in multi-concept generation.

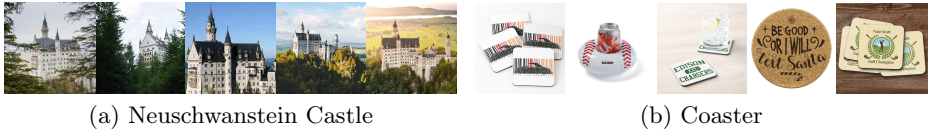


Fig. 2: Training data examples from LAION [41]: Neuschwanstein Castle exhibits minimal visual variation, while coasters appear in diverse forms and contexts.

We define the DvD phenomenon as cases where one concept (the *dominant* concept) consistently visually dominates the generation, while the other (the *dominated* concept) is strongly suppressed and rarely represented.

Illustrative example. As illustrated in Fig. 1, the prompt “Neuschwanstein Castle coaster” demonstrates this phenomenon: across multiple random seeds, the distinctive architecture of Neuschwanstein Castle dominates the generation while the coaster concept is suppressed. This pattern persists across different model versions (SD 1.4 and SD 2.1), indicating that DvD may reflect a fundamental issue in diffusion-based generation rather than a model-specific artifact.

Hypothesis: training-image visual homogeneity. We hypothesize that DvD stems from the visual homogeneity (i.e., limited variation) of concept-specific training images: concepts learned from visually homogeneous images may form relatively rigid visual priors that dominate others learned from more diverse images. To investigate this, we examine training images from the LAION dataset [41] for both concepts (Fig. 2). As shown in Fig. 2a, Neuschwanstein Castle, being a unique landmark, appears with highly consistent visual features—the iconic white facade, pointed towers, and alpine setting remain nearly identical across all training images. In contrast, Fig. 2b reveals that coasters exist in diverse forms: round plates, square tiles, and decorative pieces with various colors, patterns, and materials. These differences in training-image diversity can lead diffusion models to develop internal visual representations with different levels of flexibility.

3.2 Quantification Metric

To quantify dominance, we define the DvD Score based on whether each concept is visually present in the generated image. We focus on two-concept prompts for a direct dominant–dominated comparison. Each concept is evaluated through N binary questions using Qwen2.5-VL [2]. Let C_1 and C_2 denote the numbers of “Yes” responses for the two concepts, respectively. The DvD Score is defined as:

$$\text{DvD Score} = \frac{C_1 \times (N - C_2)}{N^2} \times 100. \quad (1)$$

This metric ranges from 0 to 100, with higher values indicating stronger dominance. We set $N = 5$ with concept-type-specific questions (e.g., for artists: “Is this image painted in the artistic style of Van Gogh?”). Appendix provides the full set of questions and per-image response examples.

3.3 DominanceBench

To systematically investigate the causes and mechanisms of DvD, we propose DominanceBench, a benchmark of 300 two-concept prompts collected from the LAION dataset [41], on which SD was trained. We focus on prompts that pair a low-diversity concept with an everyday object; the full concept lists are provided in Appendix.

Algorithm 1 DominanceBench DvD-Score-Based Prompt Filtering

- 1: Collect candidate prompts p from LAION (20–50 chars) that contain a low-diversity concept and an object.
 - 2: **for** each prompt p **do**
 - 3: Generate 10 images $\{I_1, \dots, I_{10}\}$ with SD 1.4 (different seeds).
 - 4: Compute DvD Score for each image I_i .
 - 5: Count $n = |\{i : \text{DvD Score}(I_i) > 36\}|$.
 - 6: **if** $n \geq 7$ **then**
 - 7: Add p to DominanceBench.
 - 8: **end if**
 - 9: **end for**
-

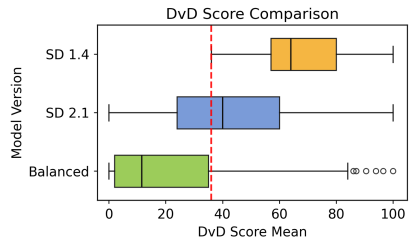


Fig. 3: Per-prompt mean DvD Scores (mean over 10 seeds) for DominanceBench (SD 1.4 and SD 2.1), with balanced prompts for reference. Red dashed line: DvD Score threshold (36).

More specifically, the 300 prompts in DominanceBench are selected by filtering of the prompts from LAION, as shown in Algorithm 1. In short, if 7 out of 10 images generated from a candidate prompt with different random seeds show a DvD Score higher than 36, the prompt is considered as a DvD prompt and included in DominanceBench. Here, the threshold is determined as follows. We build a separate set of 300 balanced two-concept prompts where both concepts successfully appear in the generated images (details in Appendix). Then, the DvD Scores are measured for the 10 images per balanced prompt with different random seeds. The third quartile of the score distribution is chosen as the threshold, as shown in Fig. 3. In the figure, it can be observed that the DominanceBench prompts have higher DvD Scores than the balanced prompts. While the initial collection is performed using SD 1.4, a substantial portion of the DominanceBench prompts still exhibit high dominance in SD 2.1 as also shown in Fig. 3.

4 Analysis

In this section, we analyze the causes and mechanisms of DvD on SD 1.4 and SD 2.1. We first validate our data-centric hypothesis via UNet fine-tuning experiments (Sec. 4.1), then analyze cross-attention dynamics (Sec. 4.2), and finally conduct a head ablation study (Sec. 4.3) to compare DvD with memorization.

4.1 Verifying the Role of Visual Homogeneity in Training Images

Takeaway 1

Lower visual diversity leads to stronger dominance. When a concept is learned from limited variations, its representation becomes overfit to specific visual patterns, causing it to dominate other concepts in multi-concept compositions.

In Sec. 3.1, we hypothesized that DvD stems from the visual homogeneity (i.e., limited variation) of concept-specific training images. We test this hypothesis through controlled fine-tuning experiments that systematically vary within-concept visual diversity.

Experimental Setup. We follow the DreamBooth setup for learning new concepts in SD fine-tuning [39]. Specifically, we fine-tune three separate models to learn “sks castle”, “sks mountain”, and “sks arena”, each defined by pairing the identifier token “sks” with a class word. We choose “sks” as a token without meaningful semantic prior, helping isolate concept learning from pre-existing semantic associations in the text encoder. To obtain concept-specific images with controllable diversity, we retrieve LAION [41] corresponding to three specific landmarks for castle, mountain, and arena, respectively—Neuschwanstein Castle, Machu Picchu, and Colosseum—whose images are visually highly similar, making it straightforward to construct highly homogeneous training sets. We use these sites as the visual targets for the three class words while keeping the training prompt generic (to avoid name-specific priors).

We fine-tune only the UNet (keeping the text encoder fixed) of SD 1.4 and SD 2.1 to learn these concepts. This setting serves as a proxy for concept learning in SD training, enabling controlled manipulation of training-image diversity. All models are trained for 100 epochs with a learning rate of 1×10^{-5} . We use LAION captions only for retrieval; to construct image-text pairs for fine-tuning, we generate a caption for each selected image using BLIP-2 [25].

Diversity Control. We construct five 120-image training sets $\mathcal{D}_k$ ($k = 1, \dots, 5$), where a smaller k indicates a more visually homogeneous image set.

For each target concept, we retrieve images whose LAION captions contain the corresponding proper name. We then construct each $\mathcal{D}_k$ by grouping and sampling images based on a cosine similarity threshold τ_k of CLIP [33] embeddings with $(\tau_1, \dots, \tau_5) = (0.95, 0.90, 0.85, 0.80, 0.75)$, where a higher τ_k yields in a more visually homogeneous set (Fig. 4a).

The controlled diversity ordering based on CLIP is additionally verified using DINOv2 [29] embeddings. Fig. 4b confirms that the overall diversity trend is consistent in both embeddings, supporting the reliability of the constructed training sets.

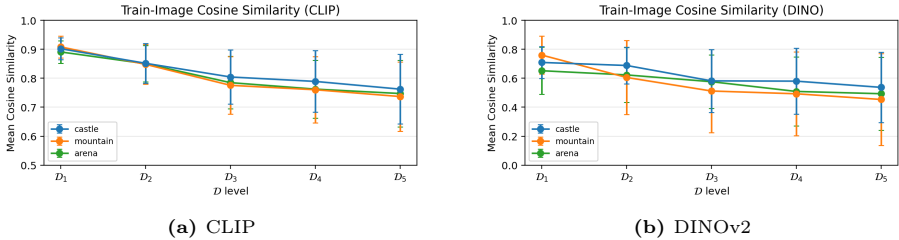


Fig. 4: Diversity control and validation for the constructed training sets. We show mean intra-set cosine similarity (mean $\pm$ std) across diversity configurations $\mathcal{D}_k$ ($k = 1, \dots, 5$), measured using CLIP (used to construct the sets) and DINOv2 (used for validation). Higher similarity indicates more visually homogeneous training images.

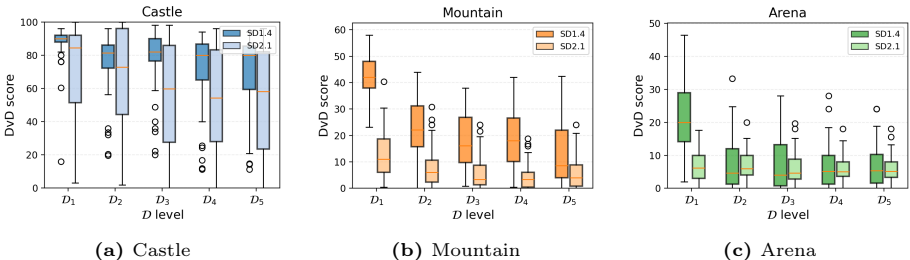


Fig. 5: Effect of visual diversity control on dominance. We show the distributions of per-prompt mean DvD Scores across different diversity configurations ($\mathcal{D}_1$ – $\mathcal{D}_5$) for the three learned concepts.

Evaluation. We evaluate dominance on 100 two-concept prompts that pair a learned concept (e.g., “sks castle”) with diverse partner concepts. We first construct 50 base prompts where the learned concept appears first, followed by the partner concept. Then, in order to eliminate concept-order effects, a flipped-order version of each base prompt is obtained, yielding 100 prompts in total. For each prompt, we generate 10 images per fine-tuned model variant using different random seeds, compute the DvD Score (Sec. 3.2), and report per-prompt mean scores, treating the learned concept as C_1 . Example evaluation prompts are provided in Appendix.

Results. Fig. 5 summarizes the effect of diversity control for the learned concepts. For both SD 1.4 and SD 2.1, the distributions of per-prompt mean DvD Scores tend to shift toward higher values as within-concept diversity decreases (smaller k in $\mathcal{D}_k$).

Collectively, these controlled experiments provide causal evidence for our data-centric hypothesis: visual homogeneity in training images exacerbates dominance in multi-concept generation (higher DvD Scores). This suggests that learning a concept from limited variations can yield overly rigid visual priors that suppress other concepts. As a qualitative illustration, Fig. 6 shows generated examples across diversity configurations for each concept.

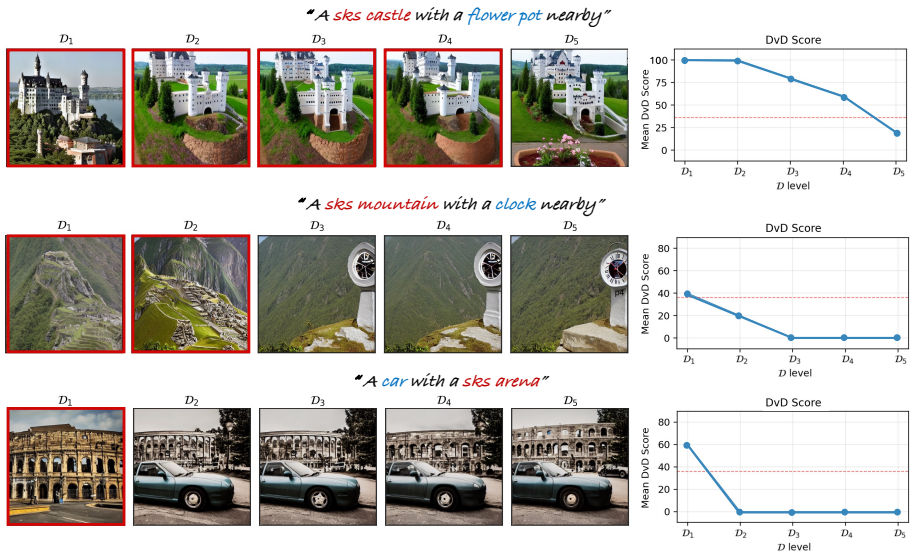


Fig. 6: Representative generation examples across diversity configurations for the three learned concepts. Images outlined in red indicate DvD cases.

4.2 Cross-Attention Dynamics in DvD

DvD can be viewed as a competition between concept word tokens in a multi-concept prompt, where one token's conditioning influence often dominates and suppresses the others. Since prompt information is injected into the UNet primarily through cross-attention, we analyze cross-attention patterns in DominanceBench prompts to localize where attention becomes imbalanced and how this imbalance evolves through denoising timesteps. We first fix the denoising timestep ($t = 50$) and scan UNet layers to identify where attention concentrates, then track how attention to the dominant and dominated tokens changes over subsequent timesteps in these layers.

Layer-wise Attention Focus at the First Denoising Step.

Takeaway 2

The dominant token in a DvD prompt tends to receive the maximum attention in low-resolution blocks already at the first denoising step.

To quantify how strongly the model focuses on specific tokens, we define a *focus score*:

$$\text{Focus}^{(\ell,t)} = \frac{\max_i a_i^{(\ell,t)} - \bar{a}_{\text{others}}^{(\ell,t)}}{H(\mathbf{a}^{(\ell,t)}) / \log_2 N + \epsilon} \quad (2)$$

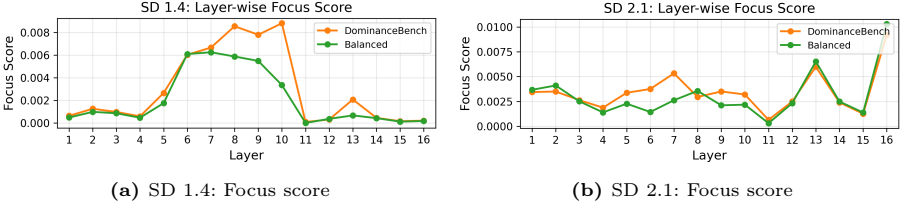


Fig. 7: Focus scores across UNet layers at the first denoising step ($t = 50$) for the DominanceBench prompts and the balanced prompts.

where $\mathbf{a}^{(\ell,t)} = (a_1^{(\ell,t)}, \dots, a_N^{(\ell,t)})$ represents the cross-attention weights over N prompt tokens at layer ℓ and timestep t (averaged across all spatial locations and attention heads), $\max_i a_i^{(\ell,t)}$ is the maximum attention weight, $\bar{a}_{\text{others}}^{(\ell,t)}$ is the mean of all other attention weights, $H(\mathbf{a}^{(\ell,t)})$ is the entropy of the attention distribution, and ϵ is a small constant for numerical stability. We provide additional motivation for this entropy-based normalization in Appendix.

Intuitively, the focus score captures whether cross-attention forms a sharp peak on a single token or is distributed evenly across the prompt. Higher values indicate stronger single-token concentration.

Experimental Setup. We compute focus scores in all UNet layers during the first denoising step ($t = 50$). Following the UNet ordering of cross-attention modules, layers 1–6 correspond to the downsampling blocks, layer 7 to the mid block, and layers 8–16 to the upsampling blocks. In particular, layers 5–10 comprise the low-resolution stage.

Results. In Fig. 7, the DominanceBench prompts show a distinctive pattern in focus scores (i.e., significantly higher focus scores) in the low-resolution stage (layers 5–10) compared to the balanced prompts. This indicates that cross-attention becomes sharply peaked already at the first denoising step, i.e., one token receives excessive attention weights.

We further examine which specific concept token is emphasized. Since layers 8–10 show a particularly large focus score gap between DominanceBench and balanced prompts in both SD 1.4 and SD 2.1, we identify the maximally attended token in these layers. In SD 1.4, the dominant concept token receives the maximum attention in 253 out of 300 DominanceBench prompts (84.33%). Similarly in SD 2.1, the dominant concept token is the most frequently maximally attended token (162 out of 300 prompts; 54.00%).

This confirms that the influence of the dominant concept token becomes amplified through the cross-attention operation in the low-resolution stage at the first denoising step. In what follows, we investigate the temporal dynamics of the influences of the dominant and dominated concepts over the subsequent denoising steps.

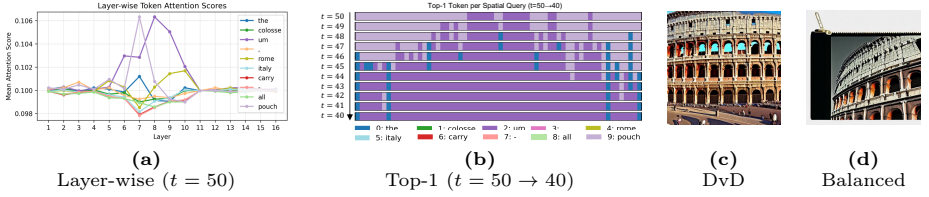


Fig. 8: Example cross-attention patterns for a DominanceBench prompt. (a) Layer-wise token attention at the first denoising step ($t = 50$). (b) Top-1 token maps across spatial locations from $t = 50$ to $t = 40$; regions where the dominated token (“pouch”) is top-1 rapidly diminish. (c) DvD outcome (dominated concept missing). (d) Balanced outcome (both concepts present).

Temporal Dynamics of Cross-Attention.

Takeaway 3

Dominated concepts rapidly lose attention in early denoising timesteps. Within the first few denoising steps, attention to the dominated token fades, while the dominant token retains a clear peak.

In some DvD prompts, the dominated token still receives substantial attention at $t = 50$, yet it can fail to appear in the final image. Fig. 8 shows one such case (“The Colosseum Rome Italy Carry-all Pouch”): at $t = 50$, the token “pouch” is strongly attended in the mid block (Fig. 8a). Yet the spatial locations where it is the top-1 token (highest-attended token per spatial location) quickly shrink from $t = 50$ to $t = 40$ (Fig. 8b), leading to a DvD outcome (Fig. 8c) rather than balanced generation (Fig. 8d). To characterize when this reduction occurs, we track how attention to the dominant and dominated tokens evolves over early denoising timesteps.

Experimental Setup. Using the dominant/dominated token labels provided by DominanceBench, we compute layer-wise attention at the first denoising step ($t = 50$) to decide where to track each token (Fig. 9). Across both models, the dominant token shows the peak in the first upsampling blocks (layers 8–10), while the dominated token peaks in the mid block (layer 7). We therefore track the dominant token in layers 8–10 and the dominated token in layer 7 over early denoising timesteps.

To quantify the temporal evolution of attention for these tokens, we define the attention deviation for token i at timestep t as $\alpha_i^{(\ell,t)} = a_i^{(\ell,t)} - \bar{a}_{\text{others}}^{(\ell,t)}$ (with the same averaging across spatial locations and heads as in Eq. (2)). Note that we use the raw attention deviation rather than the entropy-normalized focus score, since we track changes over timesteps within a single prompt (with fixed prompt length). Then, the attention change is:

$$\Delta\alpha_i^{(\ell,t)} = \alpha_i^{(\ell,t+1)} - \alpha_i^{(\ell,t)} \quad (3)$$

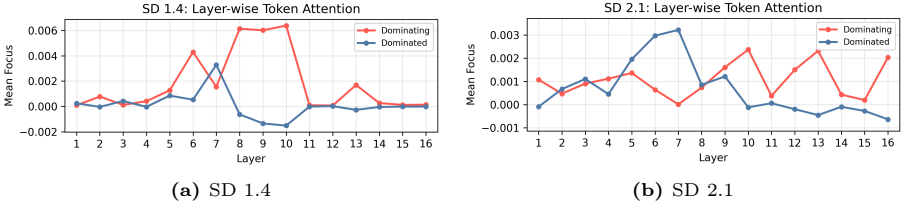


Fig. 9: Layer-wise token attention at the first denoising step ($t = 50$) for the dominant and dominated tokens (averaged over prompts).

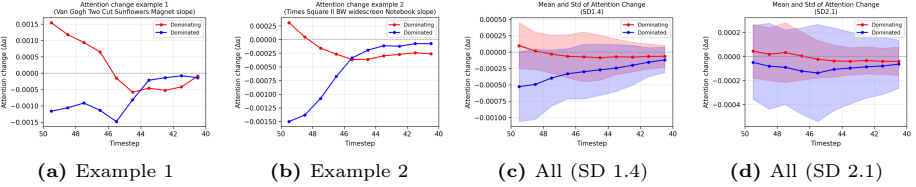


Fig. 10: Attention change ($\Delta\alpha$) across timestep intervals. (a,b) Two SD 1.4 examples. (c,d) Aggregated trends over all DominanceBench prompts on SD 1.4 and SD 2.1, respectively. The dominated token (blue) tends to show negative changes from the earliest timesteps, while the dominant token (red) maintains positive or near-zero changes initially.

In what follows, we average the attention changes over $\ell \in \{8, 9, 10\}$ for the dominant token and use $\ell = 7$ for dominated tokens. Negative $\Delta\alpha$ indicates decreasing concentration on that token.

Results. Fig. 10 summarizes attention changes ($\Delta\alpha$) across early timestep intervals. Individual prompts (Fig. 10a, 10b) show that the dominated token already has negative $\Delta\alpha$ values in the earliest timestep interval (50–40), indicating rapid attention decrease. The prompt-averaged curves over all DominanceBench prompts (Fig. 10c, 10d) show a similar pattern on both SD 1.4 and SD 2.1: the dominated token tends to stay below zero, while the dominant token remains positive or near-zero initially. This suggests that the imbalance is set up within the first few timesteps, and the dominated token loses its conditioning influence as denoising progresses.

4.3 Head Ablation Study

Takeaway 4

At the head level, our ablation results suggest that DvD is more distributed across attention heads, whereas memorization tends to be localized to a small subset of heads. Accordingly, DvD is harder to mitigate by ablating only a few heads.

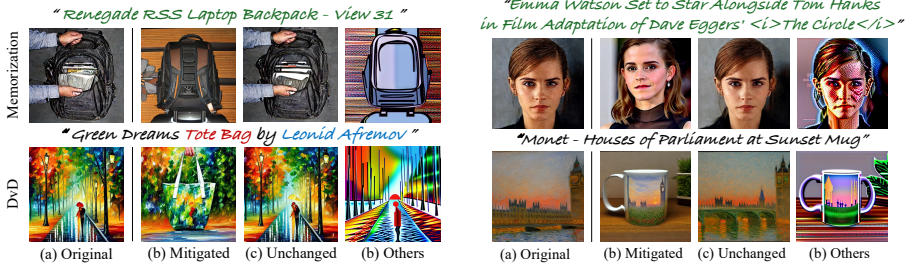


Fig. 11: Examples of ablation outcomes. Each panel shows (left to right) the unablated generation, a *Mitigated* case, an *Unchanged* case, and an *Others* case (degraded output).

Building on our cross-attention analysis, we move from layer-level patterns to individual attention heads—a practical unit of intervention within multi-head attention. We aim to answer the following question: *is DvD driven by a small set of critical heads, or does it emerge from distributed contributions across heads?*

To ground this head-level analysis, we use memorization as a reference point. Neuron-level analyses suggest that memorization can be localized to a small number of internal units [14]. Motivated by this, we use head ablation as a coarse probe to compare head-level localization between DvD and memorization, using 300 DvD prompts from DominanceBench and 500 memorized prompts identified in prior work [5], respectively.

Head Ablation Procedure. To assess each head’s contribution, we ablate target heads by scaling their attention logits (pre-softmax scores) with a small factor throughout denoising. Formally, the ablated attention logit for head h in layer ℓ is:

$$\tilde{s}^{(\ell,h)} = \begin{cases} \varepsilon \cdot s^{(\ell,h)}, & \text{if } \ell = \ell^*, h \in \mathcal{H}^*, \\ s^{(\ell,h)}, & \text{otherwise,} \end{cases} \quad (4)$$

where $s^{(\ell,h)} \in \mathbb{R}^{P \times N}$ are the attention logits for head h over P spatial queries and N text tokens (computed at each denoising step), ℓ^* is the target layer, $\mathcal{H}^* \subseteq \{1, \dots, H\}$ is the set of ablated heads, and $\varepsilon = 10^{-5}$ is a small scaling factor that effectively suppresses the head’s influence.

Outcome Classification. We classify each generation result after ablation into three categories using Self-Supervised Descriptor for Image Copy Detection (SSCD) [31], Learned Perceptual Image Patch Similarity (LPIPS) [52], and (for DvD prompts) the DvD Score (Sec. 3.2), as illustrated in Fig. 11. SSCD measures similarity to training images (lower indicates less similarity / less copying in our usage). LPIPS measures perceptual distance (higher indicates larger visual change).

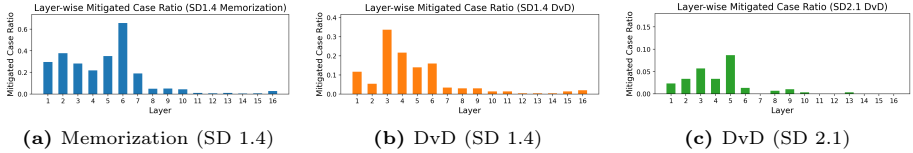


Fig. 12: Layer-wise ratio of mitigated cases in single-head ablation. For each layer, we report the proportion of the prompts that can be mitigated by ablating *any* head in that layer. Mitigation effects concentrate primarily in layers 1–6.

- **Mitigated:** The target phenomenon (memorization or DvD) is successfully reduced.
 - *Memorization:* $\text{SSCD} < 0.5$ and $\text{LPIPS} > 0.6$
 - *DvD:* $\text{LPIPS} > 0.5$ and $\text{DvD Score} < 36$
- **Unchanged:** The ablated output remains similar to the unablated generation and the target phenomenon persists.
 - *Memorization:* $\text{SSCD} \geq 0.5$ or $\text{LPIPS} \leq 0.6$
 - *DvD:* $\text{LPIPS} \leq 0.5$ or $\text{DvD Score} \geq 36$
- **Others:** The generated image quality degrades substantially (e.g., severe artifacts or incoherent structure).

Single-Head Ablation. We first ablate individual heads by setting $|\mathcal{H}^*| = 1$ in Eq. (4). For each prompt $p \in \mathcal{P}$ (where $\mathcal{P}$ contains 300 DominanceBench prompts or 500 memorization prompts), we test all heads in every layer (1–16), i.e., all $h \in \{1, \dots, H\}$ for every layer of the model.

We consider a prompt mitigated if ablating any head (in any layer) yields a *Mitigated* outcome. Under this criterion, single-head ablation mitigates 388 out of 500 memorization prompts on SD 1.4 (77.6%), but only 136/300 (45.3%) on SD 1.4 and 36/300 (12.0%) on SD 2.1. As shown in Fig. 12 mitigation concentrates in layers 1–6, i.e., the UNet downsampling blocks, with limited effects in higher layers.

Multi-Head Ablation. Next, we conduct multi-head ablation for $|\mathcal{H}^*| = 2$ and 3 in order to examine whether the DvD phenomenon is localized in a few heads or is distributed across a larger number of heads. We focus the multi-head ablation on the downsampling blocks, where single-head ablation shows high mitigation rates (Fig. 12). As enumerating all head pairs/triplets quickly becomes expensive (a layer with H heads has $\binom{H}{2}$ possible pairs and $\binom{H}{3}$ possible triplets), we instead randomly sample 10 head pairs ($|\mathcal{H}^*| = 2$) and 10 head triplets ($|\mathcal{H}^*| = 3$) per layer. For each prompt p and layer ℓ , we ablate the heads in each sampled set simultaneously following Eq. (4) and classify the resulting images as *Mitigated*, *Unchanged*, or *Others*. We compute the outcome proportions over the sampled sets and average them over prompts to obtain the layer-wise outcome proportions (which sum to one).

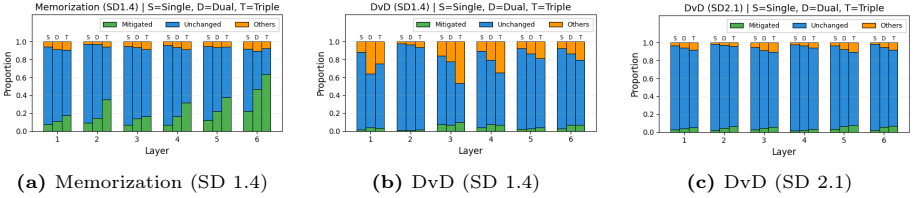


Fig. 13: Multi-head ablation outcome proportions across layers 1–6 as the ablation size increases (S=single, D=dual, T=triple).

Fig. 13 summarizes how outcome proportions change as we ablate more heads within the same layer. For memorization (SD 1.4), increasing the ablation size mainly shifts outcomes from *Unchanged* to *Mitigated* while keeping *Others* low, consistent with head-level localization. For DvD, however, mitigation improves little as we increase the ablation size. On SD 1.4, outcomes increasingly shift to *Others* rather than *Mitigated*, meaning that ablation of more heads mainly causes degraded outputs with little increase in *Mitigated*. On SD 2.1, most cases remain *Unchanged* even under dual/triple ablations. In sum, these patterns demonstrate that the DvD phenomenon is attributed to contributions spread across many heads rather than a small set of critical heads.

5 Conclusion

This work analyzed the DvD phenomenon in text-to-image diffusion models and introduced DominanceBench (300 prompts) for systematic study. We provided causal evidence that visual homogeneity in concept-specific training images amplifies dominance by inducing rigid visual priors. Mechanistically, DvD arises from sharply peaked cross-attention in low-resolution blocks at the first denoising step and an early decrease of the dominated token’s conditioning influence. Head ablations further suggested that, unlike memorization, DvD is supported by the contributions of many attention heads; ablating more heads tends to degrade the outputs without reliably reducing dominance.

These results point to a practical data-curation lever: monitoring and reducing concept-level visual redundancy (near-duplicates or highly homogeneous images) during dataset curation, and encouraging diverse reference sets in customization. Finally, the proposed focus score provides a lightweight generation-time diagnostic at the first denoising step, enabling early detection and debugging (see Appendix for additional results).

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Dominant vs. Dominated: Concept-Level

Generative Collapse in Diffusion Models

Supplementary Material

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S1 DominanceBench

S1.1 Training-Image Visual Homogeneity Analysis

DominanceBench (Sec. 3.3) categorizes artists, landmarks, and characters as high-homogeneity concepts and common objects as low-homogeneity concepts based on the training-image visual homogeneity hypothesis. Here, we provide empirical support for this categorization by analyzing the relative visual homogeneity of LAION training images for each concept in Table S1

Table S1: Concept sets used to collect DominanceBench prompts.

Category Terms	
Artist	Amedeo Modigliani, Andy Warhol, Camille Pissarro, Caravaggio, Claude Monet, Edgar Degas, Edvard Munch, Egon Schiele, Frida Kahlo, Gustav Klimt, Hokusai, Kandinsky, Leonardo da Vinci, Leonid Afremov, Monet, Pierre-Auguste Renoir, Raphael, Rembrandt, Titian, Van Gogh, Vermeer
Character	Barbie, Batman, Black Panther, Buzz Lightyear, Captain America, Cinderella, Goku, Hello Kitty, Hulk, Iron Man, Joker, Luffy, Mickey Mouse, Minions, Naruto, Pikachu, Rapunzel, Simpson, Spider-Man, Superman, Thor, Wonder Woman, Woody
Landmark	Acropolis, Big Ben, Brandenburg Gate, Colosseum, Eiffel Tower, Empire State Building, Golden Gate Bridge, Grand Canyon, Great Wall of China, Hagia Sophia, Kremlin, Machu Picchu, Neuschwanstein Castle, Parthenon, Petra, Pyramid, Space Needle, Taj Mahal, Times Square, Tower Bridge
Object	Apron, Bag, Battery Charger, Bottle, Canvas Bag, Carry-All Pouch, Clock, Coaster, Coffee Mug, Cup, Curtain, Halloween, Headphones, Hoodie, Lamp, Magnet, Magnets, Mouse Pad, Mousepad, Mug, Necklace, Notebook, Placemat, Pouch, Rug, Sleeves, Spiral Notebook, Sweatshirt, T-Shirt, Tattoo, Tie, Tote Bag, Umbrella, Wallet

For each concept, we keep the top 1,000 most frequent LAION captions containing the keyword and download the corresponding images. We then compute CLIP ViT-L/14 image embeddings and measure intra-category cosine distances to assess relative visual homogeneity.

Fig. S1 shows the cosine distance distribution for each category. We report medians to reduce the influence of near-duplicate samples. Landmark images exhibit the lowest median distance (0.3079), indicating the highest visual homogeneity. Character images follow (median 0.4417), while artist images (median 0.4660) and object images (median 0.4672) display substantially higher intra-category distances, indicating lower visual homogeneity.

Overall, this analysis provides evidence of a relative ordering in training-image visual homogeneity, with landmark images showing the highest homogeneity and object images the lowest.

S1.2 Prompt Collection and Evaluation Details

DominanceBench prompts rely on four concept sets: $\mathcal{A}$ (artist), $\mathcal{C}$ (character), $\mathcal{L}$ (landmark), and $\mathcal{O}$ (object). Table S1 enumerates their elements, and we denote their union by $\mathcal{V} = \mathcal{A} \cup \mathcal{C} \cup \mathcal{L} \cup \mathcal{O}$. We use these sets to retrieve candidate prompts from LAION and filter them using the DvD-Score-based procedure in Algorithm 1

To compute DvD Scores, we use category-specific questions for each concept type. The five questions per category are listed in Table S2

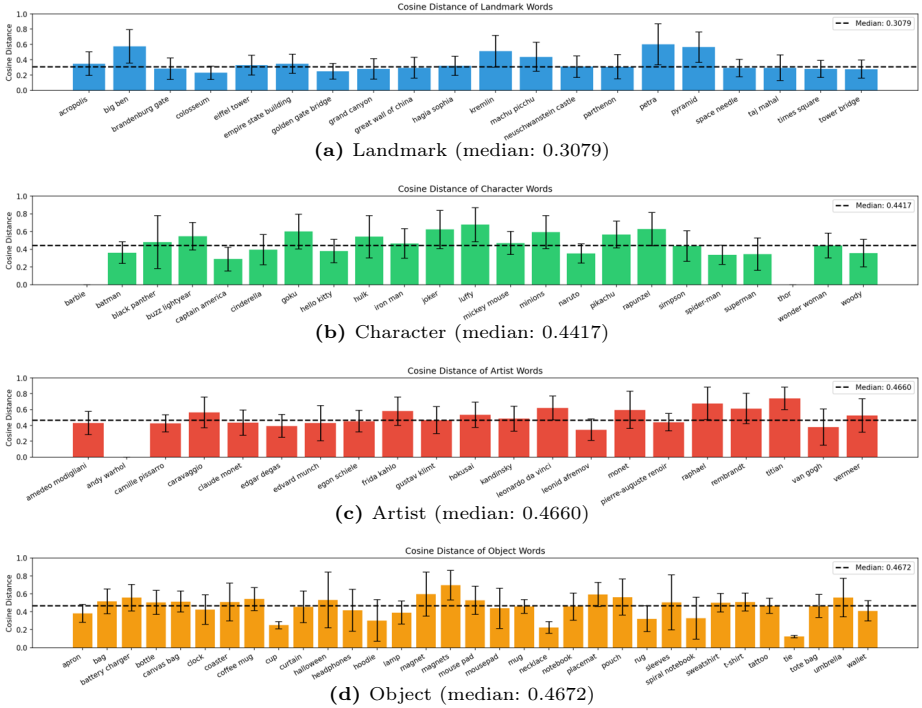


Fig. S1: Intra-category cosine distance distributions for LAION training images. For each concept in Table S1, we collect the training images associated with the 1,000 most frequent LAION captions containing the keyword and compute cosine distances in CLIP space. Lower distances indicate higher visual homogeneity.

S2 Reference Prompts

S2.1 Balanced Prompts

In Sec. 3.3 and Sec. 4.2, we use balanced prompts to distinguish DvD behavior from successful multi-concept generation.

We leverage the non-memorized benchmark of Ren et al. [36], a reference set used in their memorization comparison, which contains 500 prompts describing common subjects and objects. For each prompt, we ask GPT-5 [S2] to identify the major subjects or objects that should be visually represented in the generated image. We retain only prompts for which it identifies exactly two concepts, yielding 300 prompts. This filtered subset matches the two-concept structure of DominanceBench.

Because the balanced prompt set contains diverse common concepts, we evaluate it using the generic questions in Table S3 about whether each concept is present, rather than the category-specific templates in Table S2. We therefore use its DvD Score distribution primarily as a reference distribution for threshold calibration, as shown in Fig. 3, rather than as a strictly matched absolute benchmark against DominanceBench.

Table S2: Questions used in collecting DominanceBench for each category.

Category	Questions
Artist	– Is this image painted in the artistic style of Artist , using realistic lighting and smooth brush strokes?
	– Does the person or subject look painted with the realistic portrait technique typical of Artist ?
	– Does this image show brushwork, color tones, or shading similar to paintings by Artist ?
	– Does this look like a painting created by Artist , rather than a photo or modern artwork?
	– Is the visual composition, lighting, and realism clearly similar to Artist 's paintings?
Landmark	– Does the image show architectural details or a skyline resembling the Landmark ?
	– Is the Landmark the main subject or part of the background in this image?
	– Is the Landmark or part of it visible anywhere in the image?
	– Is this a photograph or realistic depiction of the Landmark ?
Character	– Is the structure clearly identifiable as the Landmark ?
	– Does the image contain a logo, costume, or any reference to Character ?
	– Is the scene related to, inspired by, or set in the world of Character ?
	– Is there a visual element or theme in the image connected to Character ?
	– Does the person or figure look like Character ?
Object	– Is Character visible in the image?
	– Does this image look like a product photo or include a depiction of a Object ?
	– Is there a Object included or partly shown in the image?
	– Is the appearance or shape recognizable as a Object ?
	– Does the image feature or focus on a Object ?
	– Is a Object visible in the image?

Table S3: Questions used in calculating DvD Score for balanced prompts.

Category	Questions
Concept	– Does the image include Concept as part of the scene?
	– Does the image show Concept in a recognizable way?
	– Is Concept visually represented in the image?
	– Can you see Concept clearly in the image?
	– Is Concept present in the image?

S2.2 Memorized Prompts

In Sec. 4.3, we compare DvD with memorization using 500 memorized prompts identified by Carlini et al. [5]. Among existing studies on memorization, Wen et al. [48] proposed a detection method that computes the L2 norm of text-conditional noise predictions at the first denoising step. A higher L2 norm value indicates that the prompt consistently reproduces the same image regardless of the random seed. To further characterize DominanceBench, we apply this metric to three prompt sets: memorized prompts, DominanceBench, and balanced prompts.

Fig. S2 shows the L2 norm distributions for these three prompt sets. Memorized prompts show the highest values, DominanceBench shows intermediate values, and balanced prompts show the lowest. This intermediate position suggests that DominanceBench shares some generation-time characteristics with memorization.

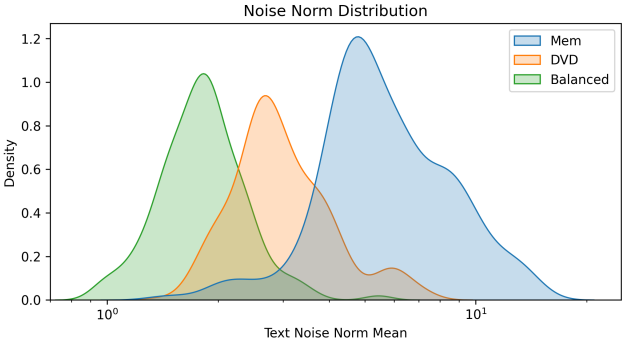


Fig. S2: L2 norm distributions of text-conditional noise predictions at the first denoising step across memorized, DominanceBench, and balanced prompts.

Table S4: Representative prompt examples used in the visual homogeneity experiments.

Class	Mode	Partner Concept	Prompt
castle	normal	fountain	A sks castle with a fountain nearby
	flipped	fountain	A fountain next to a sks castle
mountain	normal	flag	A sks mountain with a flag nearby
	flipped	flag	A flag next to a sks mountain
arena	normal	statue	A sks arena with a statue nearby
	flipped	statue	A statue next to a sks arena
minions	normal	street lamp	A sks minions with a street lamp nearby
	flipped	street lamp	A street lamp next to a sks minions
van gogh	normal	bench	A bench in the style of sks van gogh
	flipped	bench	In the style of sks van gogh, a bench

S3

Visual Homogeneity Experiments: Details and Extensions

S3.1

Prompt Examples

In Sec. 4.1, we evaluate the learned concept (e.g., “sks castle”) together with a partner concept in two-concept prompts. We include both a *normal* order and a *flipped* order for each concept pair, and compute DvD Scores over 10 random seeds. Table S4 lists representative prompt examples used for evaluation.

S3.2

Additional Concept Types

To examine whether the trend observed in Sec. 4.1 extends beyond the landmark-based concepts, we repeat the same controlled homogeneity experiment on two additional concept types: the character concept *Minions* and the artist-style concept *Van Gogh*.

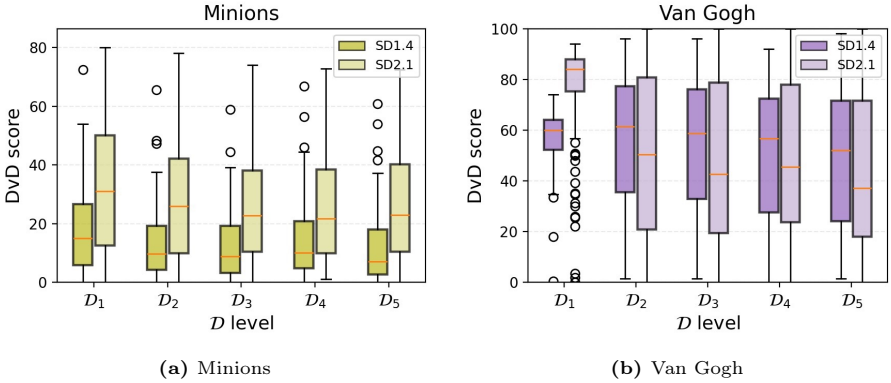


Fig. S3: Effect of training-image visual homogeneity on dominance for two additional concept types, *i.e.*, a character *Minions* and an artist-style concept *Van Gogh*. For both SD 1.4 and SD 2.1, we show the distributions of per-prompt mean DvD Scores across homogeneity configurations. Higher scores in more homogeneous settings indicate stronger dominance.

Following Sec. 4.1 for each concept we construct five 120-image training sets, D_k ($k = 1, \dots, 5$), where smaller k denotes a more visually homogeneous set. For each D_k , we fine-tune only the UNet of SD 1.4 and SD 2.1 while keeping the text encoder fixed. We also use the same prompt construction, normal/flipped concept orders, and DvD Score computation as in Sec. 4.1. For *Minions*, we evaluate whether the generated image contains both the learned character and the partner concept. For *Van Gogh*, we evaluate whether the generated image preserves the learned style together with the partner concept. Figure S3 shows the distributions of per-prompt mean DvD Scores across homogeneity configurations for the two additional concepts. For both SD 1.4 and SD 2.1, the score distributions for *Minions* and *Van Gogh* shift toward higher values as training-image visual homogeneity increases. This trend is most pronounced in the most homogeneous settings (D_1), where the learned concept more frequently dominates the partner concept. In contrast, less homogeneous settings yield more balanced generations.

These results suggest that the relationship between visual homogeneity and dominance is not limited to the landmark-based concepts used in the main paper, but also extends to character and artist-style concepts.

S3.3 Additional VLM Evaluator

To test whether the conclusion in Sec. 4.1 depends on the choice of VLM evaluator, we recompute DvD Scores using InternVL2.5-8B [S1] in place of Qwen2.5-VL while keeping the generated images, prompts, homogeneity configurations, and the questions used to assess whether each concept is present unchanged. This experiment tests the robustness of the observed relationship between training-image visual homogeneity and dominance to the choice of VLM evaluator.

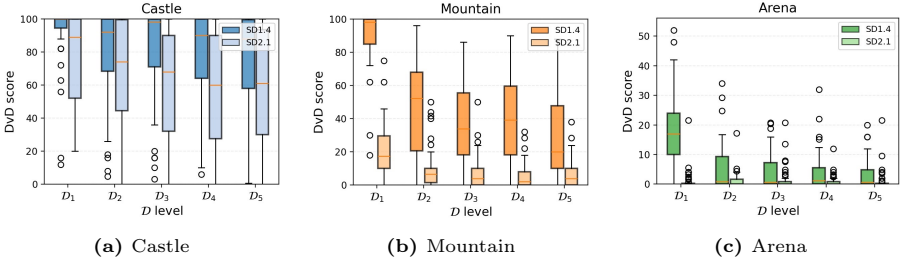


Fig. S4: DvD Score distributions recomputed with InternVL2.5-8B as the VLM evaluator for the three learned concepts.

Experimental Setup. We reuse the generated images from the homogeneity-controlled training sets $\mathcal{D}_k$ ($k = 1, \dots, 5$) for the three learned concepts: *castle*, *mountain*, and *arena*. For each image, we use InternVL2.5-8B to reassess the visual presence of the two concepts and compute the DvD Score in the same manner as Fig. 5. As in Sec. 4.1, we average the scores over 10 random seeds and report the distributions of per-prompt mean DvD Scores separately for SD 1.4 and SD 2.1.

Results. Figure S4 shows that the InternVL2.5-8B-based scores follow the same qualitative trend as Fig. 5. For both SD 1.4 and SD 2.1, the DvD Score distributions tend to shift toward higher values as training-image visual homogeneity increases (smaller k in $\mathcal{D}_k$). Overall, this evaluation preserves the central conclusion that greater visual homogeneity is associated with stronger dominance.

S4 Dominant-Token Detection in Other Layers

In Sec. 4.2, we analyze dominant-token signals in SD 2.1 using layers 8–10. This choice follows the layer-wise trend observed in Fig. 7, which shows that the gap in Focus Scores between DominanceBench and balanced prompts peaks around layers 8–10 in both SD 1.4 and SD 2.1.

However, when viewed in isolation, Fig. 7b may give a different impression: the apparent gap between the Balanced and DvD cases appears larger in layers 5–7 than in layers 8–10. To better understand where dominant-token signals are most consistently captured, we conduct an additional layer-wise analysis.

For each layer (and layer group), we measure how frequently dominant tokens are detected across 300 prompts. Tab. S5 summarizes the results. When analyzed individually, layer 10 yields the highest detection rate (72.0%), indicating that dominant-token signals are most clearly detected at this layer. Because layers 5 and 6 correspond to the last downblocks, we additionally report their grouped detection rate. When grouping layers, the average detection rate of layers 8–10 (54.0%) is substantially higher than that of layers 5–6 (26.00%).

Overall, while the last downblocks (layers 5–6) may show larger differences between the Balanced and DvD cases in Fig. 7b, dominant-token signals are more consistently detected when using layers 8–10. For this reason, the SD 2.1 analysis in Sec. 4.2 focuses on layers 8–10.

Table S5: Layer-wise detection rate of dominant tokens in SD 2.1 across 300 prompts. “Included” denotes the number of prompts in which the dominant concept token is identified as the maximally attended token in the corresponding layer.

Layer Setting	5	6	7	8	9	10	5–6	8–10	5–10
Included	103	71	30	103	137	216	78	162	98
Percent	34.33%	23.67%	10.00%	34.33%	45.67%	72.00%	26.00%	54.00%	32.67%

S5 Metric Design for Temporal Analysis

While the Focus Score (Eq. 2) uses entropy normalization, our temporal analysis (Sec. 4.2) uses only attention deviation without entropy. We explain why below.

S5.1 Different Objectives Require Different Metrics

The Focus Score compares attention patterns across different prompts with varying characteristics. The numerator measures attention concentration on the peak token relative to others, while the denominator uses entropy to capture the overall dispersion of the attention distribution. Normalizing entropy by $\log_2 N$ accounts for prompt length, enabling fair comparison across prompts with different token counts.

In contrast, Temporal Analysis tracks attention dynamics within a single prompt over time. The attention deviation $\alpha_i^{(\ell,t)} = a_i^{(\ell,t)} - \bar{a}_{others}^{(\ell,t)}$ already captures relative token importance, and its temporal change $\Delta\alpha_i^{(\ell,t)}$ directly measures how the competitive balance shifts between concepts. We do not use entropy normalization because it causes distortion from irrelevant tokens, as explained below.

S5.2 Entropy Causes Noise in Temporal Analysis

Entropy H measures the dispersion of the entire attention distribution, including tokens irrelevant to the dominant and dominated concepts (denoted as C_D and C_d , respectively). When irrelevant tokens shift attention, entropy changes even if the attention to C_D and C_d remains unchanged, leading to distorted interpretations of concept-level dynamics.

As a concrete example, consider 10 tokens (C_D , C_d , and 8 other tokens) with $\sum a_i = 1.0$ (Table S6):

Table S6: Attention weights at consecutive timesteps. The attention of C_D and C_d remains constant while that of an irrelevant token T_1 suddenly increases.

Token	Timestep t	Timestep $t + 1$	Change
C_D (Dominant)	0.40	0.40	0
C_d (Dominated)	0.30	0.30	0
T_1 (Another token)	0.05	0.15	+0.10
Other 7 tokens	0.036 each	0.021 each	-0.015 each

Using attention deviation (our approach):

$$\bar{a}_{others}^{(t)} = \frac{1 - 0.40}{9} \approx 0.067 \quad (5)$$

$$\bar{a}_{others}^{(t+1)} = \frac{1 - 0.40}{9} \approx 0.067 \quad (6)$$

$$\alpha_{C_D}^{(t)} = 0.40 - 0.067 = 0.333 \quad (7)$$

$$\alpha_{C_D}^{(t+1)} = 0.40 - 0.067 = 0.333 \quad (8)$$

$$\Delta\alpha_{C_D} = 0 \quad (9)$$

Correct interpretation: C_D 's relative advantage is unchanged.

If entropy normalization is used hypothetically:

At timestep t , attention is concentrated primarily on C_D (0.40) and C_d (0.30), with other tokens having small weights (e.g., 0.05, 0.036 each), yielding low H_t . At timestep $t + 1$, the attention of T_1 increases to 0.15, while those of C_D (0.40) and C_d (0.30) remain constant, yielding higher H_{t+1} .

An entropy-normalized metric would give:

$$\text{Score}_t = \frac{\alpha_{C_D}^{(t)}}{H_t / \log_2 N} \quad (\text{high value}) \quad (10)$$

$$\text{Score}_{t+1} = \frac{\alpha_{C_D}^{(t+1)}}{H_{t+1} / \log_2 N} \quad (\text{lower value}) \quad (11)$$

$$\Delta\text{Score} < 0 \quad (12)$$

Distorted interpretation: “ C_D 's dominance decreased”—despite α_{C_D} remaining constant at 0.40. This distortion occurs because the entropy increase stems from irrelevant token T_1 (0.05 $\rightarrow$ 0.15), not from changes in the dominant (C_D) or dominated (C_d) concepts that Temporal Analysis aims to isolate.

S5.3 Additional Layer-Wise Consideration

Beyond the token-level noise discussed above, incorporating entropy would create an additional issue for cross-layer tracking. As described in Sec. 4.2, we track dominant concept tokens in layers 8–10 and dominated tokens in layer 7. Since each layer has its own entropy value, any metric involving entropy cannot be directly compared across layers. Attention deviation $\alpha_i^{(l,t)} = a_i^{(l,t)} - \bar{a}_{others}^{(l,t)}$ isolates relative token importance within each layer, enabling consistent cross-layer comparison.

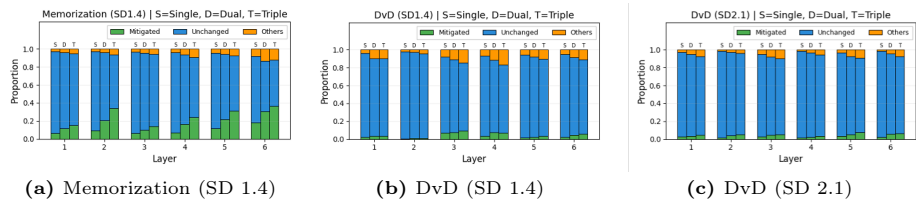


Fig. S5: Multi-head ablation outcome proportions across layers 1–6 under early-timestep ablation (timesteps 50 to 40).

Summary. By using only attention deviation, our Temporal Analysis (1) maintains direct interpretability ($\Delta\alpha$ = change in relative advantage), (2) eliminates noise from irrelevant tokens, and (3) enables consistent cross-layer tracking of the dominant and dominated concepts that characterize the DvD phenomenon.

S6 Multi-Head Ablation: Across Early Timesteps

In Sec. 4.3, we ablate specific heads throughout the full denoising process to examine whether each phenomenon is localized to a small set of heads or distributed across many heads. However, Sec. 4.2 shows that the attention imbalance underlying DvD is already established within the first few denoising steps. Here, we check whether the same head-level pattern remains observed when the analysis is restricted to the early denoising steps only.

We repeat the multi-head ablation protocol in Sec. 4.3 over early timesteps. The only change is that ablation is applied from timesteps 50 to 40. As in Sec. 4.3 we evaluate single-, dual-, and triple-head ablations in layers 1–6 and classify the resulting outputs as *Mitigated*, *Unchanged*, or *Others*.

Fig. S5 shows that the qualitative contrast between memorization and DvD remains similar even in this early-timestep setting. For memorization, increasing the ablation size mainly shifts outcomes from *Unchanged* to *Mitigated*, which is consistent with a more localized head-level mechanism. In contrast, for DvD on SD 1.4, increasing the ablation size yields only a marginal increase in *Mitigated*; the more noticeable change is a shift from *Unchanged* to *Others*. This suggests that stronger ablation tends to degrade generation quality instead of reliably reducing dominance. DvD on SD 2.1 shows the same overall tendency: *Unchanged* remains dominant across most layers and ablation sizes, while *Mitigated* cases remain rare. Overall, these results are consistent with the view that, even in the early denoising steps, DvD is supported by distributed contributions across heads rather than by a small set of critical heads. Accordingly, perturbing only a few heads is usually insufficient to mitigate DvD, and stronger ablations often harm image quality rather than removing the dominance pattern.

Table S7: Detection rates (% of prompts identified as DvD) across Focus Score thresholds and layer configurations in layers 8–10 (DominanceBench: $N_{\text{DvD}} = 300$; balanced: $N_{\text{balanced}} = 300$).

(a) SD 1.4								(b) SD 2.1							
thr	max_mean	L8	L9	L10	8&9	8&10	9&10	thr	max_mean	L8	L9	L10	8&9	8&10	9&10
DominanceBench (%)								DominanceBench (%)							
0.004	88.33	71.00	71.33	72.67	74.00	75.00	79.33	0.002	92.33	79.00	68.33	55.00	77.67	70.67	67.00
0.005	82.67	56.67	61.33	67.00	63.00	70.00	71.33	0.003	65.67	33.33	46.33	37.33	41.67	37.00	46.00
0.006	78.33	50.33	49.33	59.67	50.67	60.00	65.33	0.004	45.00	15.33	32.67	26.00	24.33	21.33	28.00
Balanced (%)								Balanced (%)							
0.004	70.33	60.00	48.33	29.67	57.00	50.67	42.33	0.002	99.00	98.67	36.67	36.33	90.67	88.00	35.00
0.005	60.33	48.67	40.00	21.33	46.67	34.67	31.67	0.003	80.33	70.33	15.33	21.67	34.00	32.67	19.33
0.006	49.33	38.00	32.00	15.00	35.67	23.67	25.00	0.004	43.00	30.33	8.67	13.33	9.00	8.67	9.33

S7 DvD Phenomenon Detection

This section explores whether our analysis results can be used for DvD detection during generation. In particular, we show that the Focus Score can serve as an early-step diagnostic for detection. While the main contribution of this paper is identifying the causes and mechanisms of the DvD phenomenon, from a practical perspective, early detection of DvD during generation can serve as a foundation for future mitigation research.

S7.1 Detection Methodology

We use the Focus Score (Eq. 2) to detect DvD. When the Focus Score exceeds a specific threshold, we determine that the prompt is likely to trigger DvD. Specifically, we compute the Focus Score in lower-resolution layers at the first denoising step ($t = 50$), and treat the token receiving maximum attention as the detected dominant concept token.

S7.2 Selected Detection Configuration

Based on Sec. 4.2 (Fig. 7) showing that DominanceBench prompts exhibit particularly high Focus Scores in layers 8–10, we consider these lower-resolution layers and tune the detection threshold. For SD 1.4 and SD 2.1, we sweep the threshold from 0.002 to 0.006 (step 0.001) and measure detection rates (% of prompts identified as DvD) on DominanceBench ($N_{\text{DvD}} = 300$) and balanced prompts ($N_{\text{balanced}} = 300$).

Table S7 reports detection rates under several layer configurations in layers 8–10. It appears that using layers 9 and 10 yields the best detection performance, *i.e.*, the maximal difference between correct detection for DominanceBench prompts and false detection for Balanced prompts (40.33%p with threshold 0.006 for SD 1.4 and 32.00%p with threshold 0.002 for SD 2.1).

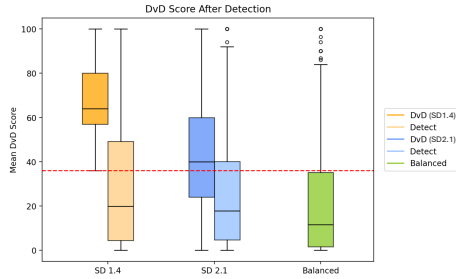


Fig. S6: DvD Score distribution before (darker boxes) and after (lighter boxes) replacing detected tokens with generic category names. Results are shown for DominanceBench prompts on SD 1.4 (orange) and SD 2.1 (blue), with balanced prompts (green) for reference. The observed score reduction suggests that the detected token actually contributes to dominance. (Red dashed line: DvD threshold of 36.)

S7.3 Validation of Detection Accuracy

To verify whether our detection method accurately identifies dominant concept tokens, we observe changes in DvD Score after modifying the detected tokens. Specifically, we apply the chosen thresholds (SD 1.4: 0.006; SD 2.1: 0.002) to DominanceBench prompts, identify the token receiving maximum attention in layers 9&10, and modify it.

Based on the training-image visual homogeneity hypothesis revealed in Sec. 4.1, we replace detected dominant tokens with generic category names such as:

- Artist: Van Gogh → “artist”
- Landmark: Colosseum → “landmark”
- Character: Spider-Man → “character”

If detection is accurate, this modification should reduce the DvD Score by replacing specific, high-homogeneity concepts with generic, low-homogeneity terms that allow more flexible representations.

Fig. S6 shows that replacing the detected dominant token substantially reduces DvD Scores on both SD 1.4 and SD 2.1, supporting accurate detection.

S7.4 Limitations and Future Work

The purpose of this section was to validate the accuracy of the detection method, and we clarify that prompt modification itself may not be a practical solution. When a user requests “Van Gogh coaster,” generating “artist coaster” undermines the original intent. Future research should explore architectural modifications or training-level methods that alleviate DvD without modifying prompts. The detection method presented in this section can be used as a diagnostic tool for developing such mitigation methods.

Supplementary References

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2. Singh, A., Fry, A., Perelman, A., Tart, A., Ganesh, A., El-Kishky, A., McLaughlin, A., Low, A., Ostrow, A., Ananthram, A., et al.: Openai gpt-5 system card. arXiv preprint arXiv:2601.03267 (2026)